

Translation of mRNA to produce proteins is highly regulated. For example, one factor that regulates eukaryotic translation is the polyadenylate binding protein (PABP). PABP upregulates translation by binding the poly-A tail of an mRNA molecule (ie, the 3' end of mRNA consisting of multiple adenosine residues) to help recruit the ribosome.

PABP consists of a single polypeptide chain and contains four distinct RNA recognition motifs (RRM's), each of which contains four β -strands and two α -helices. Mutation experiments revealed that each RRM binds poly-A tails through residues in several β -strands. Crystallography showed that the two α -helices in each RRM, which consist of amino acids 9-24 and 52-60, respectively, help hold the β -strands in the correct orientation for RNA binding.

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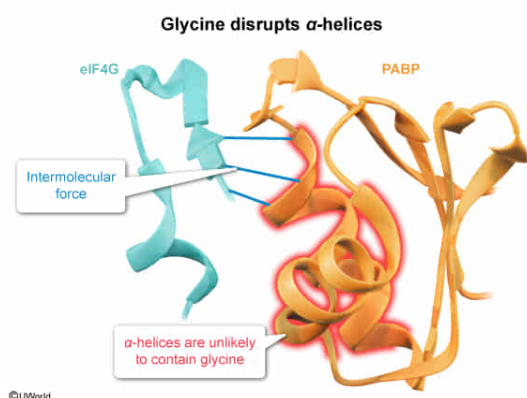
Each of the following amino acid residues is likely to be found in the portion of the RRM that binds eIF4G EXCEPT:

- ☐ A. Leucine [8%]
- ☐ B. Arginine [28%]
- ✓ ☒ C. Glycine [52%]
- ☐ D. Serine [9%]

Correct

53% answered correctly

Explanation:



Protein **secondary structure** consists primarily of α -helices and β -strands (the latter of which are often organized into β -sheets). Each of these structures are stabilized by hydrogen bonds between backbone amide groups, where the N-H group of one **amide** interacts with the C=O group of another amide. Additionally, each form of secondary structure has a characteristic

repeating pattern of bond angles. Disruption of these bond angles can destabilize the secondary structure.

The passage states that each RRM region of PABP binds to eIF4G through residues found in α -helices. Two amino acids are known to disrupt α -helices: glycine and proline. Neither of these amino acids is likely to be found within an α -helix. Therefore, of the given answer choices, **glycine is least likely** to be found in the eIF4G-binding portion of an RRM region.

(Choices A, B, and D) Both polar and nonpolar amino acids are commonly found in α -helices. Leucine and other hydrophobic residues are often found in α -helices at protein-protein interfaces or in transmembrane regions, whereas polar or charged amino acids such as serine or arginine are often found in portions of α -helices that are exposed to an aqueous environment.

Educational objective:

Proline and glycine disrupt α -helices and are not commonly found within them.

Time spent:0 sec

Subject:
Biochemistry

QID:403258

System:
Amino Acids and Proteins

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Given that PABP is primarily found in the reducing environment of the cytosol, the information in the passage best supports the conclusion that the RNA-binding β -strands are held in the correct orientation for binding by:

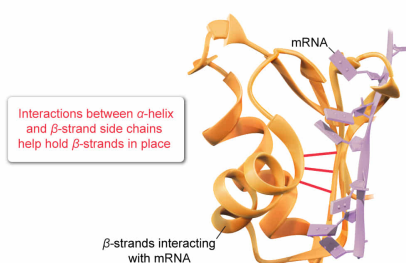
- ☐ A. covalent bonds between amino acid side chains. [19%]
- ☐ B. covalent bonds between amino acid backbone groups. [16%]
- ✓ ☒ C. noncovalent interactions between amino acid side chains. [39%]
- ☐ D. noncovalent interactions between amino acid backbone groups. [24%]

Correct

40% answered correctly

Explanation:

Tertiary structure is stabilized by interactions between side chains



The **tertiary structure** of a protein is its three-dimensional, folded form. Tertiary structure

is **primarily stabilized** by **noncovalent interactions between side chains**, often bringing amino acids that are far apart in the primary structure closer together. Correct tertiary structure is essential for most proteins to carry out their biological functions.

The passage states that each RRM region binds mRNA through residues in several β -strands, which are partially held in place by interactions with two α -helices. These helices consist of amino acids 9-24 and 52-60, respectively. Therefore, the two α -helices are far apart from each other in the amino acid sequence and must be brought into proximity with each other and the β -strands through tertiary structure interactions. Consequently, the β -strands are most likely held in the proper orientation for mRNA binding by noncovalent interactions between β -strand side chain groups and the side chains of the two α -helices.

(Choice A) The only covalent bond between amino acid side chains that is ever involved in maintaining tertiary structure is a **disulfide bond**. However, disulfide bond formation requires an oxidizing environment. The question states that PABP is primarily found in the *reducing* environment of the cytosol, so disulfide bonds are unlikely to form.

(Choice B) Covalent bonds between amino acid backbone groups (ie, the non-side group portion of amino acids) are **peptide bonds** and are the basis for **primary protein structure** (ie, amino acid sequence), not tertiary structure.

(Choice D) Noncovalent interactions between amino acid backbone groups are primarily found in **secondary structural elements**, not tertiary structural elements.

Educational objective:

Protein tertiary structure is primarily stabilized by interactions between amino acid side chains, often bringing secondary structural elements that are far apart in the amino acid sequence closer together.

Time spent:0 sec

Subject:
Biochemistry

QID:403259

System:
Amino Acids and Proteins

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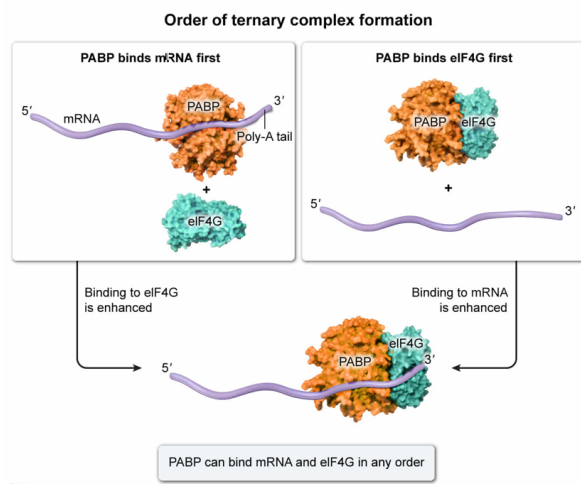
Based on the information in the passage, which of the following best describes the order of formation of the ternary complex?

- ☐ A. PABP must bind mRNA poly-A tails before it can bind eIF4G. [26%]
- ☐ B. PABP must bind eIF4G before it can bind mRNA poly-A tails. [11%]
- ☐ C. PABP binding interactions with eIF4G and mRNA poly-A tails must form simultaneously. [11%]
- ☒ D. PABP may bind mRNA poly-A tails and eIF4G in any order. [49%]

Correct

48% answered correctly

Explanation:



A **ternary complex** is a group of **three molecules bound together**, generally with at least one of the molecules being a protein. Ternary complexes may form by either of **two mechanisms**: ordered, in which the ligands must bind in a specific sequence, or random, in which the order of binding does not matter.

The complex consisting of PABP, eIF4G, and mRNA is an example of a ternary complex. According to the passage, PABP affinity for mRNA poly-A tails increases 10-fold when eIF4G is added. This suggests that some amount of binding occurs in the absence of eIF4G (if no binding occurred in the absence of eIF4G, the increase in binding affinity would be *infinite* upon eIF4G addition, not 10-fold). Similarly, PABP affinity for eIF4G increases 12-fold when mRNA with a poly-A tail is added, suggesting that PABP can bind eIF4G in the absence of any mRNA. Based on this information, **PABP may bind mRNA poly-A tails and eIF4G in any order.**

(Choices A and B) These choices are examples of *ordered* binding mechanisms. If PABP needed to bind mRNA poly-A tails prior to binding eIF4G, no eIF4G binding would be observed in the absence of mRNA. Similarly, no mRNA binding would be observed in the absence of eIF4G if the PABP-eIF4G complex needed to form first.

(Choice C) Simultaneous binding of eIF4G and mRNA by PABP would require that all three molecules collide at the same time in the correct orientation. Mechanisms of this nature are extremely unlikely and rare. Additionally, this mechanism is not supported by the information in the passage, which shows that either mRNA or eIF4G can be bound to PABP in the absence of the other.

Educational objective:

Ternary complexes are groups of three molecules (at least one of which is usually a protein) bound together. These complexes may form in a particular order or in random order, depending on the specific molecules involved.

Time spent: 0 sec
 Subject:
 Biochemistry

QID: 403260
 System:
 Amino Acids and Proteins

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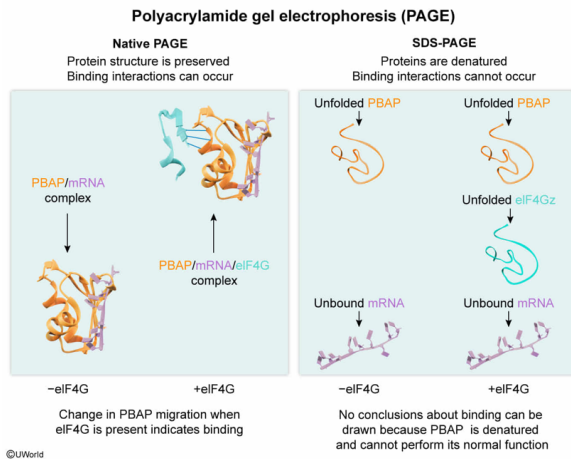
Which of the following best explains why researchers used native gels for the electrophoresis experiments?

- ✔ ☒ A. To preserve the three-dimensional structure of the proteins involved [79%]
- ☐ B. To ensure that all proteins of the same mass migrated the same distance [4%]
- ☐ C. To confer approximately equal charge-to-mass ratios on all proteins assessed [9%]
- ☐ D. To separate PABP into its individual domains so each could be assessed independently [6%]

Correct

78% answered correctly

Explanation:



Gel electrophoresis is a technique used to separate and analyze proteins and other biomolecules. Proteins are most commonly analyzed by polyacrylamide gel electrophoresis (PAGE), which can be performed under various conditions:

- **Native PAGE** takes place in a polyacrylamide gel in the absence of any detergents or reducing agents. These gels preserve the native structure of the protein, allowing for analysis of interactions with other molecules.
- **SDS-PAGE** is PAGE performed in the presence of sodium dodecyl sulfate (SDS), a detergent that causes the protein to denature (ie, unfold) and coats it with negative charge. This version of electrophoresis separates proteins on the basis of mass only. This approach is useful for analyzing changes in protein molecular weight as a result of post-translational modifications, or for simply visualizing the different protein sizes present in a mixture. Reducing agents (eg, dithiothreitol) may be added to SDS-PAGE experiments to break any disulfide bonds that are present.

In the given experiment, researchers were monitoring interactions between PABP and its binding partners (mRNA poly-A tails and the protein eIF4G). In general, **protein binding interactions can occur only when the proteins are folded correctly**. Therefore, all proteins present in the described experiments needed to maintain their three-dimensional structures. **This can only occur in native gels.**

(Choice B) In native gels, proteins of the same mass do not necessarily migrate the same distance. Differing net charges and protein shapes commonly impact protein migration through native gels.

(Choice C) Proteins have approximately equal charge-to-mass ratios in SDS-PAGE, *not* native PAGE. This is what allows SDS-PAGE to separate proteins on the basis of size only.

(Choice D) **Protein domains** are separate regions of a single polypeptide chain that fold independently and perform distinct functions. Because they are all part of the same polypeptide (ie, the same primary structure), neither native nor SDS-PAGE can separate protein domains. (Note: SDS-PAGE may separate protein **subunits**, but not domains within a subunit.)

Educational objective:

Native polyacrylamide gel electrophoresis preserves the 3D folded forms of proteins, allowing interactions between proteins and other molecules to occur within the gel.

Time spent:0 sec

Subject:
Biochemistry

QID:403261

System:
Amino Acids and Proteins

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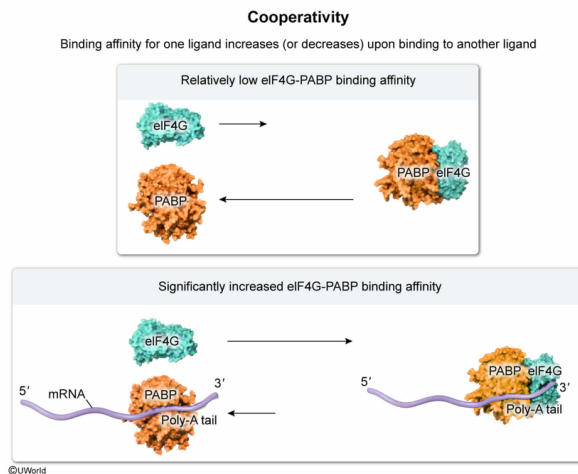
Which of the following descriptions of the interactions between PABP, eIF4G, and the poly-A tail of mRNA is best supported by the passage?

- ☐ A. Both PABP and eIF4G bind mRNA poly-A tails directly to increase translation levels. [19%]
- ☐ B. eIF4G binding to the complex attenuates the effect of PABP binding. [16%]
- ☒ C. Formation of the ternary complex exhibits cooperativity. [61%]
- ☐ D. mRNA translation does not occur in the absence of PABP or eIF4G. [2%]

Correct

61% answered correctly

Explanation:



Binding cooperativity occurs when the **binding of one ligand** to a protein **increases (or decreases) the affinity of that protein for another ligand** at a separate binding site. Such changes in affinity result exclusively from **conformational changes**: the protein changes shape upon the binding of one ligand, and this shape change allows the protein to bind another ligand more (or less) easily. This phenomenon commonly occurs in proteins that bind multiple molecules of the same ligand, but it can also occur for the binding of two or more different ligands.

The passage states that the affinity of PABP for mRNA poly-A tails increases 10-fold when the protein eIF4G is added. Similarly, the affinity of PABP for eIF4G increases 12-fold when mRNA with poly-A tails is added. Both these observations suggest that the binding of PABP to either of the ligands induces a conformational change that enhances binding of the other ligand. Therefore, **formation of the ternary complex** between PABP, eIF4G, and mRNA **exhibits cooperativity**.

(Choice A) The passage indicates that eIF4G binds to PABP and that PABP binds mRNA poly-A tails, but there is no information to indicate the eIF4G interacts directly with poly-A tails.

(Choice B) The passage states that PABP binding to poly-A tails upregulates translation. It further indicates that formation of the ternary complex enhances translation beyond what is achieved with PABP alone. Therefore, eIF4G *increases* the effect of PABP rather than attenuating (ie, decreasing) it.

(Choice D) Although the passage indicates that both PABP and eIF4G help upregulate the translation of mRNA, it does *not* indicate that translation is completely abolished in the absence of these proteins.

Educational objective:

Binding cooperativity occurs when the binding of a protein to one ligand induces a conformational change that alters the protein's affinity for another ligand at a separate binding site.